

.NET Software Engineer

Technical Assessment

Owner: Fourth Recruitment Sofia **Role:** .NET Software Engineer

Overview

This assessment is designed to give you space to demonstrate how you think through a business problem and deliver a working solution.

Please use the [Microsoft Northwind sample database](#) as your data source and build your solution using the Microsoft .NET stack (C# / .NET).

Business Scenario

Northwind Traders is a fictional company that sells food products to customers worldwide. The business needs a simple internal tool that allows staff to look up customers and review their order history.

Your task is to build a back-end service that exposes this data in a clean, well-structured way.

What the Application Should Do

Customer Overview

Staff should be able to retrieve a list of customers; with each customer's name and the number of orders they have placed. It should be possible to search or filter by customer name.

Customer Detail

When a specific customer is selected, the service should return the customer's details along with a summary of their order history. For each order, the following should be available:

- The total value of the order
- The number of products included in the order

Structure of Your Solution

We leave the architecture entirely to you. Here are a few things to keep in mind:

- If you do build a front-end, it should communicate with the back end exclusively over HTTP — it should not have direct access to the database layer.
- You may structure the solution with as many projects or layers as you feel is appropriate.

Testing

Ideally, include at least one automated test. There is no requirement for full coverage — we are simply interested in how you think about testability and what you consider worth verifying.

Submission

Please upload your project to a public [GitHub](#) or [GitLab](#) repository. Include a README that explains:

- How to run your project locally
- Any assumptions or trade-offs you made
- Anything you would improve or do differently if you have more time

A Note on AI Tools

You can use AI tools. If you do, please include your prompts.

What We Are Looking For

We value clear thinking, clean code, and good judgement. We are as interested in the choices you make and the reasoning behind them as we are in the code itself.

A simple, focused, well-reasoned solution will impress us more than a complex one.